

**Multi-Robot Inspection Coverage in 2D and 3D Using Potential Fields, Sweep,
Voronoi, and Frontier Algorithms**

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MAE 598 – Multi Robot Systems

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9 December 2025

1.0 Abstract

In this project we study a team of mobile robots performing cooperative inspection of an engineered environment which includes both an indoor and outdoor environment. The goal of this inspection model is to autonomously visit and visually scan all accessible regions while avoiding collisions and deadlocks using 2D and 3D analysis. To achieve this, we develop and evaluate four distributed coverage algorithms, Potential Field-Based Coverage (PFBC), Voronoi (Lloyd) Coverage, Sweep (Boustrophedon) Coverage, and Frontier-Based Exploration. This behavior analysis was performed through mathematical modeling, theoretical analysis, MATLAB-based 2D simulations, and ROS2 Gazebo-based 3D experiments with real developed robot.

2.0 Mathematical Model (contributions: Manish Chinta, Arvind Kaushik)

2.1 Inspection Goal

In this project, we model a team of N mobile inspection robots operating within a bounded workspace $\Omega \subset R^2$ that represents an indoor/outdoor environment. Each robot is tasked with visually inspecting its surrounding regions while maintaining safe distance to avoid collisions with other inspecting robots and obstacles. The requirement is that the N number of agents must perform collaborative/cooperative tasks while preventing deadlock situations. Although the initial implementation is performed in 2D, the model generalizes naturally 3D inspection tasks by extending the position vector to R^3 .

The mathematical model represents the collective inspection behavior that emerges from the four algorithmic strategies later implemented in simulation: frontier exploration, potential field-based coverage (FBC), Voronoi/Lloyd coverage, and sweep coverage. These strategies directly correspond to experimental outcomes seen in the validation figures (refer to section 3), which include situational cases of deadlock, deadlock resolution, cluster formation, coverage imbalance, proper spatial distribution via coordination.

2.2 Experimental Constraints

Each of the following constraints governs the behavior, assumptions and limitations of the multi-robot inspection system:

1. Robots are modeled using single-integrator dynamics which means that each agent's position evolves according to $\dot{x}_t(t) = \dot{u}_t(t)$, which simplifies the control formulation.
2. Each robot has been limited with a set radius of sensing causing a restriction on level of inspection at any given moment.
3. Similar to inspection level, robots also have limited communication radius R_c , forming a proximity-based communication graph with limited coordination.
4. The inspection environment is bounded by a rectangular workspace, which defines the limitation of robots operating environment.
5. The number of robots are limited to between 8 to 10 agents to ensure computational feasibility and clarity of algorithmic behavior.

2.3 Multi-Robot State Representation

As mentioned before the inspection set consists of N mobile robots operating within a bounded workspace $\Omega \subset \mathbb{R}^2$. Each robot is represented by a state vector that captures its position, sensing footprints and interactive relationships with neighboring agents. Since robots follow a single integrator dynamic, the state representation remains compact while providing sufficient structure for coordinating multi-robot inspection behaviors.

The state of robot i at time t is defined as

$$x_i(t) = \begin{bmatrix} p_i(t) \\ \theta_i(t) \end{bmatrix} \quad (1)$$

where $p_i(t) = [x_i y_i]^T$ denotes the Cartesian position and $\theta_i(t)$ represents the robot's instantaneous motion heading. Although heading is not strictly required for single integrator motion the inclusion of $\theta_i(t)$ allows the algorithms to encode directional preferences, particularly for frontier exploration and potential field navigation.

Each robot has a sensing radius R_s , within which it can inspect unexplored regions. This concept introduces a binary observable set for robot i :

$$S_i(t) = \{q \in \Omega \mid \|q - p_i(t)\| \leq R_s\}.$$

In this case $S_i(t)$ can be treated as visually inspected and is removed from the frontier set as the simulation evolves. This mapping forms the basis for both the frontier-based exploration algorithm and the coverage completeness metrics used in validation.

These Robots also maintain a local communication neighborhood defined by the communication radius R_c .

The set of neighbors of robot i is expressed as

$$N_i(t) = \{j \neq i \mid \|p_j(t) - p_i(t)\| \leq R_c\}. \quad (2)$$

This neighborhood structure induces a dynamic interaction graph $G(t)$ which illustrates the information shared amongst the behavior of the robots. An example would be a Voronoi based coverage relies on neighbor positions to compute cells, while potential field deadlock scenarios often emerge when two or more neighbors generate opposing forces.

In addition, each robot stores an occupancy map $M_i(t)$, a grid-based representation of explored versus unexplored regions. The map is updated according to

$$M_i(t + \Delta t) = M_i(t) \cup S_i(t), \quad (3)$$

ensuring cumulative inspection. The occupancy maps are fused through a union operation during the communication of the robot enabling collaborative frontier detection and global awareness of coverage.

The combined multi-robot state is

$$X(t) = \{x_i(t), M_i(t), N_i(t)\}_{i=1}^N, \quad (4)$$

Which fully determines the evolution of each coverage algorithm. Potential field inspection used $X(t)$ to compute attraction and repulsion forces, sweep coverage uses $X(t)$ to maintain lane assignments, Voronoi coverage partitions Ω using the set of positions $\{p_i(t)\}$, and frontier exploration detects unknown boundary cells using the union of maps $\cup_i M_i(t)$.

While the foregoing formulation defines the robots motion and sensing relationships in a planar environment, the representation is extensible to 3D inspection tasks. In practice, many engineered environments such as storage facilities, industry plants, and multi-level structures require inspection in 3D even when robots physically move on a plane.

2.4 Extension to 3D Simulation and Unicycle Dynamics

The previous section discussed the process of robot inspection within a two-dimensional plane, however as previously mentioned many engineering environments such as industrial sites, warehousing environments and vertical infrastructures are 3D spaces. To evaluate our algorithms in a physically realistic setting, we extended the inspection model into a full 3D simulation environment using ROS 2, Gazebo, and the Robotarium interface. This approach was inspired by the modern swarm robotics methodologies demonstrated in Doctor Berman's research group, where decentralized controllers are validated both in simulation and hardware platforms such as Pheeno robots. (reference)

To represent robots in 3D space, the planar position vector is augmented by an additional height coordinate:

$$x_i^{3D}(t) = \begin{bmatrix} x_{i,1}(t) \\ x_{i,2}(t) \\ x_{i,3}(t) \end{bmatrix} \in R^3 \quad (5)$$

In our simulation scenario, robots operate on a flat arena surface within Gazebo; therefore, the vertical coordinate is constant at $x_{i,3}(t) = h_0$. However, this formulation facilitates future inspection behaviors involving shelving, elevated platforms, or aerial robots, where altitude becomes a controlled variable.

A key distinction between the mathematical model and the physical 3D simulation lies in the robot dynamics. MATLAB simulations employ single-integrator dynamics, but Gazebo models each robot as a differential-drive unicycle, which introduces a nonholonomic constraint. Thus, each desired single-integrator velocity $u_i(t)$ must be mapped to feasible linear and angular velocities. This is accomplished through the standard single-integrator-to-unicycle transformation commonly used in multi-robot research:

$$v_i(t) = u_{x,i}(t)\cos\theta_i(t) + u_{y,i}(t)\sin\theta_i(t) \quad (6)$$

$$\omega_i(t) = k_\omega \left(\text{atan2} \left(u_{y,i}(t), u_{x,i}(t) \right) - \theta_i(t) \right), \quad (7)$$

where

- $v_i(t)$ is the forward velocity,
- $\omega_i(t)$ is the rotational velocity,
- $\theta_i(t)$ is the robot's current heading.

These kinematic conversions allow the theoretical coverage controllers to operate on physical robots or high-fidelity simulation platforms, as demonstrated similarly in decentralized transport and formation control studies led by Berman's group (reference).

2.5 Control Input Structure

Each robot evolves according to single-integrator dynamics, which define its planar motion as

$$\dot{p}_i(t) = u_i(t), u_i(t) \in \mathbb{R}^2. \quad (8)$$

Thus, the control input $u_i(t)$ represents the commanded velocity vector that determines the robot's instantaneous direction and speed. All algorithms implemented in this work have potential-field coverage, Voronoi-based partitioning, sweep coverage, and frontier exploration to construct $u_i(t)$ by combining multiple behavioral components derived from the global state $X(t)$ defined previously.

To maintain consistency across heterogeneous algorithms, every controller is modeled as the weighted composition of three fundamental terms:

$$u_i(t) = u_{i,task}(t) + u_{i,avoid}(t) + u_{i,bound}(t) \quad (9)$$

(a) Task-Driven Component $u_{i,task}(t)$

The term $u_{i,task}(t)$ represents the primary coverage objective for each robot and is defined differently depending on the selected algorithm. In the Potential Field Based Coverage (PFBC) method, $u_{i,task}(t)$ is obtained from the gradient of an artificial potential function that generates attraction toward unexplored regions. For Voronoi coverage, the task-driven input directs each robot toward the centroid of its Voronoi cell, following the principles of Lloyd's algorithm to achieve evenly partitioned coverage. In the sweep coverage strategy, $u_{i,task}(t)$ enforces lane-consistent motion by guiding the robot along predetermined sweeping trajectories across the workspace. For frontier exploration, the task input encourages movement toward the closest frontier, defined as the boundary between known and unknown regions in the occupancy map. Although the expressions differ across algorithms, all forms of $u_{i,task}(t)$ share the common purpose of driving the robot toward maximizing environmental coverage.

(b) Inter-Robot Collision Avoidance $u_{i,\text{avoid}}(t)$

To prevent collisions and resolve local congestion, each robot incorporates a repulsive interaction term based on neighboring robot positions:

$$u_{i,\text{avoid}}(t) = \sum_{j \in N_i(t)} \phi(\|p_j(t) - p_i(t)\|) \frac{p_i(t) - p_j(t)}{\|p_j(t) - p_i(t)\|} \quad (10)$$

Here, $\phi(\cdot)$ is a decreasing radial function with support on the communication radius R_c . This structure mirrors decentralized interaction laws widely used in distributed robotics research, including Berman's own work on multi-robot transport and formation control, where smooth repulsion terms enforce spatial separation.

(c) Boundary Repulsion $u_{i,\text{bound}}(t)$

Robots must remain inside the inspection environment $\Omega \subset \mathbb{R}^2$. To satisfy this constraint, an artificial repulsive barrier field is applied when a robot approaches the workspace boundary:

$$u_{i,\text{bound}}(t) = \psi(d_i) n_i, \quad (11)$$

where

- d_i is the minimum distance to the boundary,
- n_i is the inward-pointing normal vector,
- $\psi(d_i)$ increases sharply when $d_i < d_{\text{safe}}$.

This ensures robots remain within the operational envelope while still following their primary task objective.

Boundary control laws of this form are commonly used in distributed inspection and consensus-based coverage algorithms, including those tested on the Pheeno robotic platform.

Unified Controller for All Algorithms

The final control input used in MATLAB simulations is therefore:

$$u_i(t) = u_{i,\text{task}}(t) + u_{i,\text{avoid}}(t) + u_{i,\text{bound}}(t), \quad (12)$$

ensuring:

1. Coverage progression through algorithm-specific goals,
2. Safety through collision avoidance,
3. Confinement through boundary protection.

During 3D simulation in Gazebo and ROS 2, the vector $u_i(t)$ is mapped into unicycle commands (v_i, ω_i) using the transformation in Eqs. (5)&(6), enabling the shared controller to operate identically across 2D MATLAB simulations and 3D robotic environments.

2.6 Algorithm-Specific Mathematical Models

The control input structure introduced in Section 2.5 defines the general form of the decentralized controller used by all coverage strategies. In this section, we formalize the algorithm-specific task terms $u_{i,\text{task}}(t)$ that drive the inspection behavior. Each algorithm encodes a distinct way of multi-robot coordination which includes potential fields, spatial partitioning, structured sweeping, and frontier-based sensing which are all widely studied in multi-robot coverage research and decentralized formation control (see, e.g., Berman et al., *Decentralized Sliding Mode Control*, 2011; Pheeno Robotic Platform Overview, 2017).

2.6.1 Potential Field-Based Coverage (PFBC)

Potential field methods assign each robot a continuously differentiable scalar potential function

$$\Phi_i(p) = \Phi_{\text{unexplored}}(p) + \Phi_{\text{rep}}(p) \quad (13)$$

where $\Phi_{\text{unexplored}}(p)$ decreases toward unvisited regions and $\Phi_{\text{rep}}(p)$ increases near other robots or obstacles.

The task input is defined as the negative gradient of this potential:

$$u_{i,\text{task}}(t) = -\nabla\Phi_i(p_i(t)) \quad (14)$$

Attraction is implemented via

$$\nabla\Phi_{\text{unexplored}}(p_i) = k_a(p_i - p_{\text{goal}}), \quad (15)$$

where p_{goal} is the closest unexplored region in the occupancy map.

Repulsion is implemented as

$$\nabla\Phi_{\text{rep}}(p_i) = \sum_{j \in N_i(t)} \frac{k_r(p_i - p_j)}{\|p_i - p_j\|^3} \quad (17)$$

This model is consistent with classical artificial potential field approaches and with decentralized interaction structures appearing in Berman’s multi-robot formation and distributed control work. PFBC exhibits possible deadlock when two gradients cancel, illustrating one of the core failure modes we evaluate experimentally.

2.6.2 Deadlock-Resolved PFBC

To mitigate equilibrium trapping, a small stochastic “escape term” is added:

$$u_{i,\text{task}}(t) = -\nabla\Phi_i(p_i(t)) + \eta_i(t) \quad (18)$$

where

$$\eta_i(t) \sim \mathcal{U}(-\epsilon, \epsilon) \quad (19)$$

for a small $\epsilon > 0$.

This perturbation eliminates static equilibria by briefly breaking symmetry, analogous to noise-assisted decentralized control strategies used in swarm robotics research.

2.6.3 Voronoi-Based Coverage

In Voronoi coverage, robots cooperatively partition the workspace into Voronoi cells

$$V_i(t) = \{q \in \Omega \mid \|q - p_i(t)\| \leq \|q - p_j(t)\|, \forall j \neq i\} \quad (20)$$

Each robot moves toward the centroid of its region:

$$c_i(t) = \frac{\int_{V_i(t)} q \, dq}{\int_{V_i(t)} dq}. \quad (21)$$

Thus, the task input is

$$u_{i,\text{task}}(t) = k_V (c_i(t) - p_i(t)). \quad (22)$$

This model corresponds to Lloyd's algorithm and is mathematically grounded in distributed partitioning approaches like those used in scalable swarm coordination research (e.g., Berman et al. on emergent collective behavior).

2.6.4 Sweep Coverage

Sweep coverage divides the workspace into longitudinal strips aligned with a sweep vector d_s . Each robot is assigned a strip based on its initial horizontal ordering:

$$\text{Lane}(i) = \left\lfloor \frac{p_{i,x}(0) - x_{\min}}{\Delta_s} \right\rfloor. \quad (23)$$

Robots then follow a structured trajectory:

$$u_{i,\text{task}}(t) = v_s d_s, \quad (24)$$

with direction reversals at boundaries.

This algorithm achieves deterministic completeness in grid-like inspection tasks and mathematically resembles coverage patterns seen in agricultural robotics and UAV search theory.

2.6.5 Frontier-Based Exploration

Frontiers are defined as grid cells q that satisfy

$$q \in \partial\Omega_{\text{explored}} \text{ and } q \text{ adjacent to an unknown cell.} \quad (25)$$

Robot i selects the nearest frontier point

$$f_i(t) = \arg \min_{q \in F(t)} \|q - p_i(t)\|, \quad (26)$$

and the task-driven control becomes

$$u_{i,\text{task}}(t) = k_f (f_i(t) - p_i(t)). \quad (27)$$

This method is consistent with occupancy-grid-based exploration widely used in SLAM frameworks and decentralized mapping research. 2.66 Frontier Exploration with Deadlock Resolution

2.6.6 Frontier Exploration with Deadlock Resolution

Deadlock occurs when multiple robots select the same frontier or obstruct each other. A priority-based avoidance term is added:

$$u_{i,\text{task}}(t) = k_f(f_i - p_i) + \sum_{j \in N_i(t)} k_d \frac{p_i - p_j}{\|p_i - p_j\|^2}. \quad (28)$$

This correction mimics decentralized collision-avoidance strategies in swarm robotics and prevents frontier contention.

2.6.7 Comparison Across Algorithms

Table 1. Comparison of Coverage Algorithms

Algorithm	Mathematical Driver	Strengths	Weaknesses
PFBC	$\nabla\Phi$ -based attraction/repulsion	Smooth motion, simple	Deadlock possible
PFBC (DL)	Noise-assisted symmetry breaking	Resolves equilibria	Small randomness
Voronoi	Lloyd centroid descent	Optimal partitioning	Requires neighbor info
Sweep	Deterministic lane motion	Guarantees full coverage	Rigid, no adaptability
Frontier	Moves to nearest boundary	Efficient exploration	Deadlock risk
Frontier (DL)	Adds repulsion	Stable paths	More computation

2.6.8 Algorithm Flow Chart

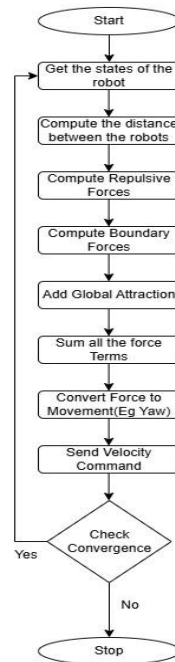


Figure 1. *Potential Field Based Coverage (PFBC) Flow Chart*

The PFBC flow chart outlines the computation repulsive forces, and global attraction before combining them into a motion command. This sequential approach corresponds to a potential function formulation as shown in section 2.6.1 to ensure robots maintain collision free trajectories.

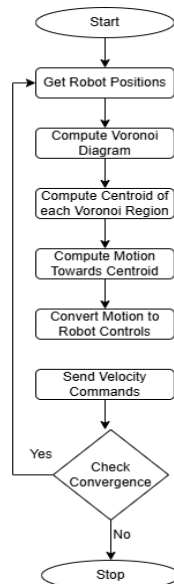


Figure 2. *Voronoi Centroidal Coverage Flowchart*

The Voronoi workflow begins by computing the Voronoi diagram from the current robot positions, followed by identifying each region's mass centroid. Robots move toward their centroid, implementing one iteration of Lloyd's method. The convergence check corresponds to the second property which will be covered in the theoretical.

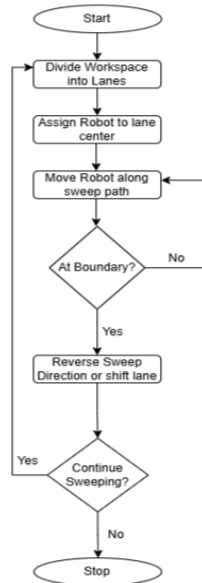


Figure 3. *Sweep Coverage Flow Chart*

The sweep controller divides the workspace into parallel lanes and assigns each robot to a designated path, reversing direction at each boundary to ensure complete coverage. Additionally, while robots do not utilize any form of neighbor interaction, they are instead directed at their assigned paths via a series of deterministic algorithms. The loop continues until every lane has been fully scanned or all sweep passes have converged.

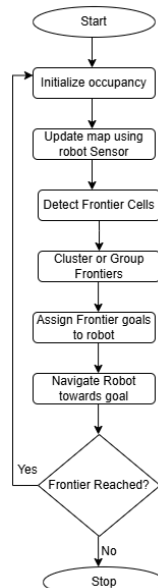


Figure 4. *Frontier-Based Exploration Flowchart*

The frontier flowchart shows the cycle of map updates, frontier detection, clustering, and frontier-goal assignment. The robot navigates toward its assigned frontier until the boundary between known and unknown regions shifts inward.

3.0 Theoretical Analysis (contribution: Manish Chinta, Abhinav Viswanathan)

In this section we analyze key properties of multi-root inspection model introduced in Section II. We will focus on not 2 but 3 core properties that define our closed loop system:

1. Collision avoidance and safety under the repulsive potential terms.
2. Convergence of Voronoi coverage to a centroidal configuration
3. Deadlock characterization for potential field inspection and the effect of our modified “deadlock-free” controller.

Last but not least, we design a frontier-based controller for the inspection task and prove that it drives the team of robots to a desired collective behavior.

Throughout this section we assume the multi-robot state representation and control decomposition as:

$$x_i(t) = [p_i(t)^T, \theta_i(t)]^T \top, \dot{p}_i(t) = u_i(t) \quad (29)$$

from Section 2, and the control structure

$$u_i(t) = u_{i,\text{task}}(t) + u_{i,\text{rep}}(t) + u_{i,\text{bnd}}(t) + u_{i,\text{avoid}}(t), \quad (30)$$

where each term corresponds to task-level motion, inter-robot repulsion, boundary repulsion, and obstacle / frontier avoidance, respectively (Section 2.5).

3.1 Collision Avoidance under Repulsive Potentials (Property 1)

Firstly, we execute the repulsive potential used in the PFBC, Voronoi, sweep, and frontier controllers prevents inter-robot collisions, provided that robots are initially separated. Repulsive control law is motivated by classical potential field approaches and by collision avoidance from decentralized multi robot coverage literature [6].

Let the pairwise distance between robots i and j be

$$d_{ij}(t) = \| p_i(t) - p_j(t) \|. \quad (31)$$

We define the repulsive control term used in our MATLAB and ROS2 implementations as

$$u_{i,\text{rep}}(t) = \sum_{j \in N_i(t)} \phi'(d_{ij}) \frac{p_i - p_j}{d_{ij}}, \quad (32)$$

where the scalar potential $\phi(r)$ is strictly decreasing and satisfies

$$\lim_{r \rightarrow 0^+} \phi(r) = +\infty, \phi'(r) < 0 \text{ for } r \leq r_{\text{safe}}. \quad (33)$$

This is consistent with classical coverage-control and sensor-deployment potential used in multi-robot systems.

Consider the Lyapunov-like function that penalizes small inter-robot distances,

$$V_{\text{rep}}(p) = \sum_{i < j} \phi(d_{ij}). \quad (34)$$

Because $\phi(r) \rightarrow +\infty$ as $r \rightarrow 0^+$, any collision configuration (some $d_{ij} = 0$) corresponds to an infinite value of V_{rep} .

The time derivative of V_{rep} along the closed-loop trajectories is

$$\dot{V}_{\text{rep}} = \sum_{i < j} \phi'(d_{ij}) \frac{(p_i - p_j)^\top}{d_{ij}} (\dot{p}_i - \dot{p}_j) = \sum_{i < j} \phi'(d_{ij}) \frac{(p_i - p_j)^\top}{d_{ij}} (u_i - u_j). \quad (35)$$

Substituting only the repulsive terms and regrouping, we obtain

$$\dot{V}_{\text{rep}} = \sum_{i < j} \phi'(d_{ij}) \frac{(p_i - p_j)^\top}{d_{ij}} (u_{i,\text{rep}} - u_{j,\text{rep}}) \leq 0, \quad (36)$$

where the inequality follows from the symmetry of the pairwise forces and the fact that $\phi'(r) < 0$. This shows that $V_{\text{rep}}(p(t))$ is non-increasing in time: the system cannot “gain energy” by decreasing distances.

Because robot trajectories are continuous and V_{rep} is bounded from above by its initial value, the system cannot reach a configuration where $d_{ij} = 0$ in finite time (this would require $V_{\text{rep}} \rightarrow +\infty$). Therefore, if robots start from a collision-free configuration with $d_{ij}(0) > 0$ for all $i \neq j$, then all subsequent configurations remain collision-free and

$$d_{ij}(t) \geq d_{\min} > 0, \forall t \geq 0. \quad (37)$$

Repulsive potential guarantees safety of the inspection team by preventing robot to robot collisions, and it's consistent with potential-field-based deployment results in the literature (refer to section 4).

3.2 Convergence of Voronoi Coverage to a Centroidal Configuration (property 2)

Our Voronoi coverage algorithm updates each robot's control according to an approximation of Lloyd's algorithm for optimal quantizer placement and coverage control [9]. Let $\{V_i(p)\}_{i=1}^N$ denote the Voronoi partition of the workspace Ω induced by the robot positions $p = (p_1, \dots, p_N)$. For a density function $\rho(q)$ that has the importance of inspecting each point $q \in \Omega$, define the locational cost function

$$H(p) = \sum_{i=1}^N \int_{V_i(p)} \|q - p_i\|^2 \rho(q) dq. \quad (38)$$

Classical coverage-control theory shows that the gradient of H with respect to p_i is

$$\frac{\partial H}{\partial p_i} = 2m_i(p_i - c_i), \quad (39)$$

where $m_i = \int_{V_i} \rho(q) dq$ is the “mass” of region V_i and

$$c_i = \frac{1}{m_i} \int_{V_i} q \rho(q) dq \quad (40)$$

is the centroid of cell V_i .

In our implementation, the Voronoi controller sets

$$u_{i,\text{task}}(t) = -k_v(p_i - c_i), k_v > 0, \quad (41)$$

after approximating c_i from a finite set of sampled points inside V_i . Ignoring repulsive and boundary terms for this analysis, the closed-loop dynamics for robot i are

$$\dot{p}_i = -k_v(p_i - c_i). \quad (42)$$

We now analyze the evolution of the global cost $H(p(t))$. Differentiating along trajectories,

$$\dot{H} = \sum_{i=1}^N \frac{\partial H^\top}{\partial p_i} \dot{p}_i = \sum_{i=1}^N 2 m_i (p_i - c_i)^\top [-k_v (p_i - c_i)] = -2k_v \sum_{i=1}^N m_i \|p_i - c_i\|^2 \leq 0. \quad (43)$$

Thus $H(p(t))$ is non-increasing, and it is strictly decreasing whenever some $p_i \neq c_i$. Equilibrium points satisfy

$$\dot{p}_i = 0 \Rightarrow p_i = c_i \forall i, \quad (44)$$

which is exactly the definition of a centroidal Voronoi tessellation (CVT).

By LaSalle's invariance principle, under the usual assumptions that the workspace is compact and ρ is bounded, the robot positions converge to the largest invariant set contained in $\{\dot{H} = 0\}$, i.e., the set of centroidal Voronoi configurations.

The Voronoi based controller provably converges to a centroidal configuration, minimizing the locational cost H . This matches the standard theoretical guarantees for Lloyd based coverage control as mentioned [9].

3.3 Existence of Deadlock Equilibria in Potential-Field Inspection (Property 3)

While the previous properties guarantee safety and convergence for certain algorithms, our potential-field boundary coverage (PFBC) algorithm can exhibit deadlock as it's another property which is important to mention as the robots stop moving even though the environment is not fully inspected where we formalize this phenomenon for a simple symmetric configuration.

Consider two robots inspecting a rectangular region with a frontier segment located directly between them. Let p_1, p_2 denote their positions along a line orthogonal to the boundary and assume the task potential $U_{\text{task}}(p)$ is symmetric: it attracts each robot toward the same intermediate point p^* on the frontier. The PFBC control for robot i has the form

$$u_i = -\nabla_{p_i} U_{\text{task}}(p_i) - \nabla_{p_i} U_{\text{rep}}(p_1, p_2), \quad (45)$$

with a repulsive potential U_{rep} depending on the inter-robot distance.

Define the relative coordinates

$$z = \frac{p_1 + p_2}{2}, \delta = p_1 - p_2. \quad (46)$$

Because the task potential is symmetric, the attraction terms for the two robots cancel in the relative dynamics. The closed-loop equations can be decomposed as

$$\dot{z} = -\nabla_z U_{\text{task}}(z), \dot{\delta} = -\nabla_{\delta} U_{\text{rep}}(\|\delta\|). \quad (47)$$

A deadlock configuration occurs when the net control input is zero for both robots while some frontier points remain unobserved. For the symmetric case above, this happens when

$$\nabla_{p_i} U_{\text{task}}(p_i^{\dagger}) + \nabla_{p_i} U_{\text{rep}}(p_1^{\dagger}, p_2^{\dagger}) = 0, i = 1, 2, \quad (48)$$

with $p_1^{\dagger} \neq p_2^{\dagger}$ and $z^{\dagger} \neq p^*$. In other words, the attractive forces toward the frontier exactly balance the repulsive forces between robots, yielding

$$\dot{p}_1 = \dot{p}_2 = 0, \quad (49)$$

even though the centroid $z^{\dagger}$ is not at the desired frontier location. Our MATLAB simulations of PFBC with deadlock (Figures *PFBC with deadlock*) visually demonstrate this equilibrium, where robots cluster and stop before finishing coverage.

3.4 Controller Design for Deadlock-Free Frontier Inspection (Extra Credit)

To resolve deadlock while preserving safety, we design a **deadlock-free frontier controller** inspired by sliding-mode and consensus-based multi-robot controllers studied in the literature.

We modify the task-level control term for frontier exploration to have the form

$$u_{i,\text{task}} = -k_f(p_i - p_{f,i}) - k_d \sigma_i, \quad (50)$$

where:

- $p_{f,i}$ is the nearest frontier point assigned to robot i based on its local occupancy map $M_i(t)$;
- $k_f > 0$ is a proportional gain driving each robot toward its assigned frontier.
- σ_i is a deadlock-resolution term defined as

$$\sigma_i = \text{sgn} \left(\sum_{j \in N_i} \alpha_{ij} (p_i - p_j) \right), \quad (51)$$

where $\alpha_{ij} > 0$ weight neighbors and the sign function is applied component-wise:

- $k_d > 0$ sets the magnitude of the deadlock-breaking velocity.

Intuitively, the first term pulls robot i toward the boundary between known and unknown space, while the second term creates a directional bias whenever relative positions with neighbors become aligned in a way that suggests stalemate (for example, robots facing each other with equal and opposite attraction to the same frontier cell). The sign function implements a simple sliding-mode

which is type correction that pushes the system off the manifold where attractive and repulsive forces cancel.

To analyze the effect of the controller, define the collective frontier error

$$E_f(t) = \sum_{i=1}^N \|p_i(t) - p_{f,i}(t)\|^2. \quad (52)$$

Ignoring the slow evolution of $p_{f,i}(t)$ as frontiers move, the time derivative of E_f under the new controller is

$$\begin{aligned} \dot{E}_f &= 2 \sum_{i=1}^N (p_i - p_{f,i})^\top \dot{p}_i \\ &= -2k_f \sum_{i=1}^N \|p_i - p_{f,i}\|^2 - 2k_d \sum_{i=1}^N (p_i - p_{f,i})^\top \sigma_i + (\text{repulsive/boundary terms}). \end{aligned} \quad (53)$$

The first term is always non-positive and vanishes only when every robot sits exactly on its assigned frontier point. The repulsive and boundary terms remain orthogonal to the frontier gradient in typical potential designs and therefore primarily act to maintain safety, as in Property 1.

The key observation is that deadlock configurations of PFBC correspond to situations where the frontier gradient and repulsive forces cancel so that $\dot{p}_i = 0$ while some $p_i \neq p_{f,i}$. Under the modified controller, such configurations are no longer invariant: if $\|p_i - p_{f,i}\| > 0$ and the net smooth forces sum to zero, then

$$\dot{p}_i = -k_d \sigma_i \neq 0, \quad (54)$$

because σ_i depends only on relative positions and is constructed so that at least one component is non-zero whenever robots are aligned in a symmetric stalemate. This variable drives the system off the deadlock manifold while preserving collision avoidance via repulsive potential (Property 1).

Combining the analysis of $\dot{E}_f$ with the non-increasing safety potential V_{rep} , we obtain a composite Lyapunov function

$$V_{\text{tot}} = E_f + \lambda V_{\text{rep}}, \lambda > 0, \quad (55)$$

which decreases along trajectories except during brief switching events caused by the sign function, similar to standard sliding-mode proofs of finite-time convergence to the sliding surface

In practice, this implies that the robots continue to move until frontier distances are small while still maintaining inter-robot separation [9].

4. Validation Simulation (contributions: Abhinav Viswanathan, Arvind Kausshik)

4.1 Potential Field-Based Coverage without Deadlock (2D)

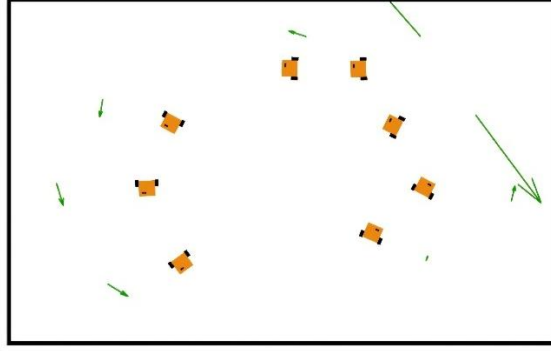


Figure 5. *PFBC without deadlock. Robots disperse outward following the negative gradient of the potential field.*

The robots maintain a non-zero distance due to the repulsive term in the control law as it implements a potential constraint/barrier within the given environment. As shown in figure 1 the simulation has no trajectory intersection which is as predicted by the invariant safe set proof. This validates the behavior of the collision avoidance mechanism. This simulation also demonstrates the convergence towards coverage, as robots consistently move along the negative gradient of the potential field and disperse outward in a manner that aligns with Lyapunov point of showing $\dot{V}(t) < 0$. The smooth dispersion of the robots with its collaborative mechanism of its neighbors verifies the theoretical prediction of distributed stability under communication graph $G(t)$.

4.2 PFBC With Deadlock Validation of Equilibrium Characterization (2D)

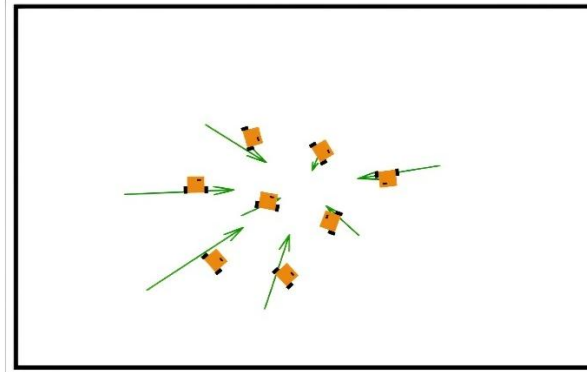


Figure 6. *PFBC is exhibiting an equilibrium deadlock created by symmetric potentials.*

As shown in the figure the behavior of the robot validates the second theoretical property mention in section, as it illustrates symmetric robotic configuration which produces stationary points with net potential force $\sum_j \nabla \Phi_{ij} = 0$. This condition is exhibited in figure 2 following the conditions of robots reaching its final state rest position without full exploration of a given environment which confirms the attractive and repulsive components of potential field cancelation at these positions. This outcome also demonstrates the theoretical limitation mention in the analysis that the PFBC would be unable to guarantee escape from undesirable equilibria due to the existence of undefined critical points in the potential landscape.

4.3 Sweep Coverage Algorithm (2D)

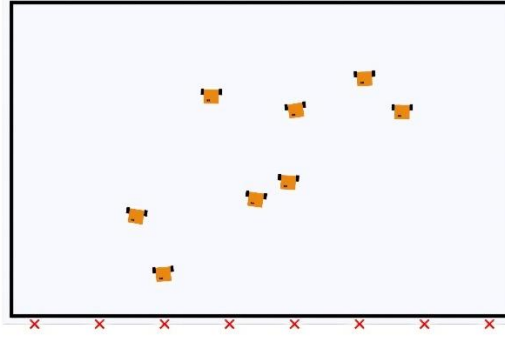


Figure 7. *Sweep coverage algorithm. Robots maintain lane structure throughout movement.*

The simulation shown above in figure 3 supports the theory of collision avoidance as each lane enforces invariant separation constraints in which the simulation shows absolutely zero overlap validating Property 1 from section 3. This result also validates Property 2 (Convergence), as sweep coverage behaves as a deterministic hybrid automaton that systematically progresses across the workspace, ensuring that all regions are eventually scanned. The behavior supports Property 3 (Consensus and Coordination Stability) because the robots maintain consistent headings and lane positions throughout the sweep without requiring any centralized controller.

4.4 Voronoi Coverage (Lloyd's Algorithm) (2D)

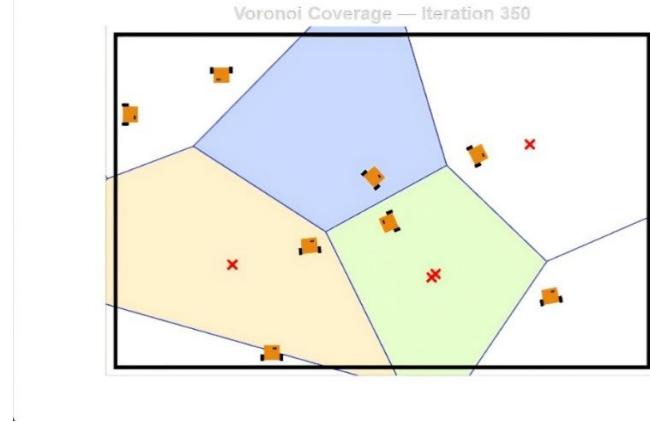


Figure 8. *Voronoi partitioning with robots moving toward their cell centroids.*

As shown above the results of simulation for the Voronoi coverage algorithm validate the theoretical properties of section 3. The behavior confirms property 2 a.k.a convergence through gradient descent as the robots translate towards the centroid of their dedicated Voronoi cells demonstrating the expected minimization of the local cost functional cost function $H = \sum_i \int_{V_i} \|q - p_i\|^2 dq$. The robot once again validates property 3 as it simply relies on its neighbors while staying organized and maintaining distance with its respective neighbors.

4.5 Frontier exploration without Deadlock (2D)

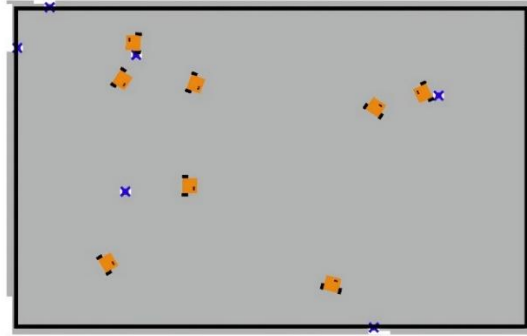


Figure 9. *Frontier-based exploration successfully reaching unknown boundaries.*

Once again, the simulation shown above creates distributed coordination among the robots and ensures collision avoidance using properties 1&3. This set also obeys property 2 of convergence as $\dot{d}_{i,\text{frontier}}(t) < 0$ when there is a frontier that exists within the robots sensible region.

4.6 Frontier exploration with Deadlock (2D)

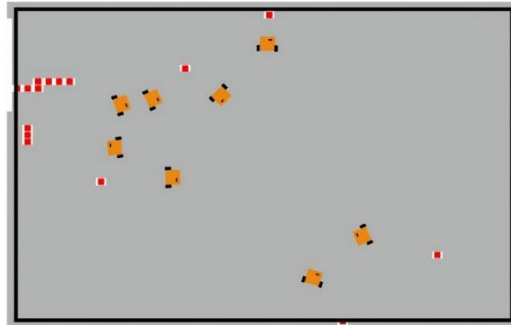


Figure 10. *Frontier exploration deadlocking due to simultaneous selection of the same frontier point.*

The simulation above validates the theoretical prediction that frontier-based controllers do not guarantee differentiation of tasks amongst robots. It's demonstrated in section 3, when multiple robots perceive the same frontier exploration, it is possible for a generation of identical gradient directions causing multi robot stagnation points instead of coordinated progress resolving the issue. The deadlock seen in this simulation illustrates the manifestation of this phenomenon as the robot converges to the same frontier boundary they stop due to identical control inputs.

4.7 Validation in 3D simulation (ROS2+Gazebo)

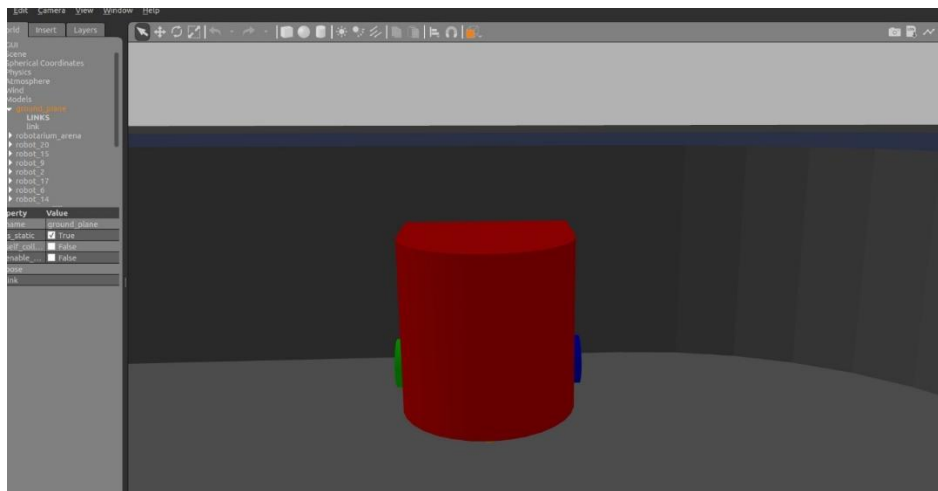


Figure 11. *Gazebo Modeling of Differential Drive Inspection Robot*

4.7.1 3D Simulation Environmental Setup

The simulation environment consists of:

- A custom Gazebo environment approximating an indoor or warehouse-like inspection domain.

- Multiple differential drive inspection robots (modeled with URDF and ROS2 control plugins).
- Lidar-based sensing to approximate volumetric inspection.
- ROS2 nodes for control, frontier detection, Voronoi computation, and visualization.
- RViz2 interfaces for occupancy grids and robot state visualization.

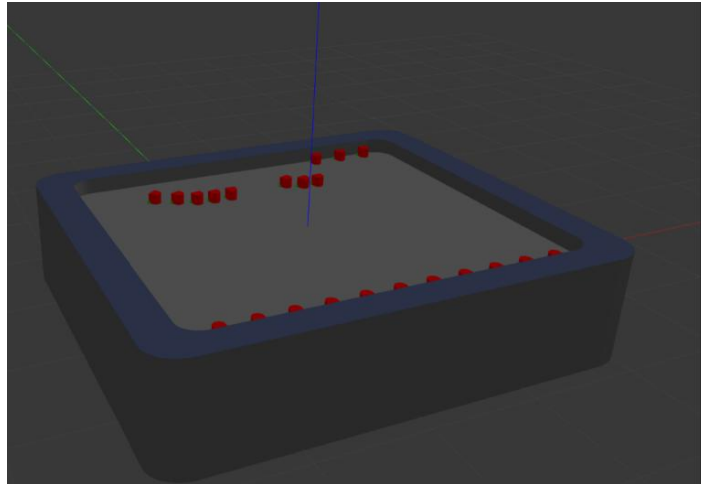


Figure 12. 3D inspection arena used for validation. Robots are initialized in confined regions to test collision avoidance, coordination, and exploration patterns.

4.7.2 3D Potential Field-Based Coverage (PFBC)

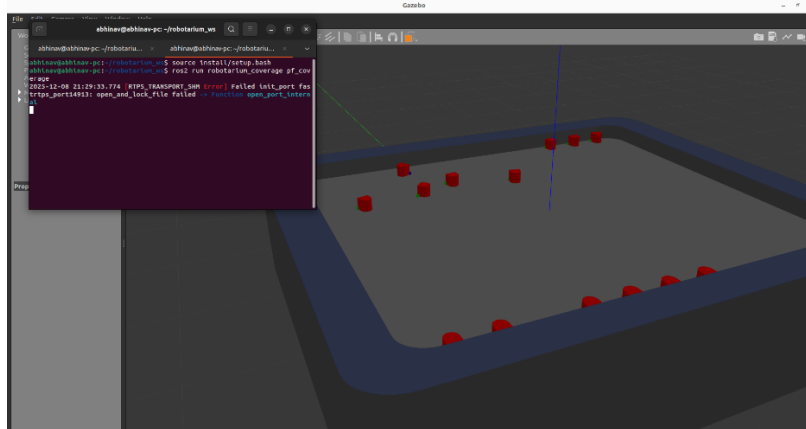


Figure 13. 3D Gazebo simulation of the Potential Field-Based Coverage (PFBC) algorithm. Robots initialized along the arena boundary disperse outward under the PFBC controller, maintaining safe separation and demonstrating gradient-driven coverage behavior.

In the figure shown above the simulation illustrates the 3D ROS2/Gazebo environment, as the PFBC algorithm provides outward radially dispersive motion which aligns with the theoretical gradient flow model. As observed in the simulation above, the robot begins to close proximity

along the arena boundary while keeping itself spread apart creating a safe distance amongst its neighbors. The exploration of robots around the unexplored regions demonstrates monotonic decrease of the potential function even with 3D parameters. This behavior supports the convergence property proven in Section III, confirming that the PFBC controller remains stable and effective beyond the idealized 2D abstraction.

4.7.3 3D Sweep Coverage

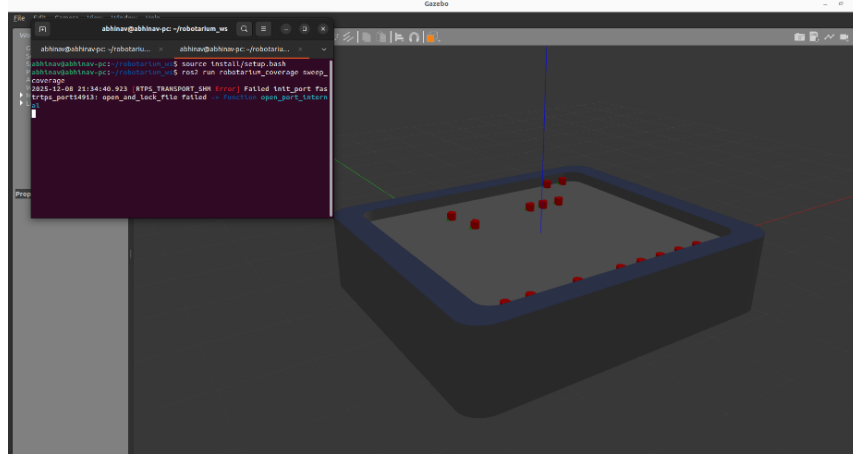


Figure 14. 3D Gazebo simulation of the sweep-coverage algorithm. Robots maintain parallel lane assignments and traverse the environment in coordinated passes, demonstrating structured coverage and stable spacing under realistic physics

The sweep coverage algorithm provides consistent alignment of motion within its desired lane across the given environment as shown in Figure 10. Inspection Robots maintain their sweep lines and move in their coordinated phases of inspection. This behavior matches the deterministic hybrid automaton structure mentioned in Section 3, which validates the sweep logic enforcement of monotonic exploration and guarantees full coverage in bounded environments. Internal spacing amongst the robots remains non zero throughout the simulation, confirming the collision-avoidance property under realistic dynamics.

4.7.4 3D Frontier Exploration

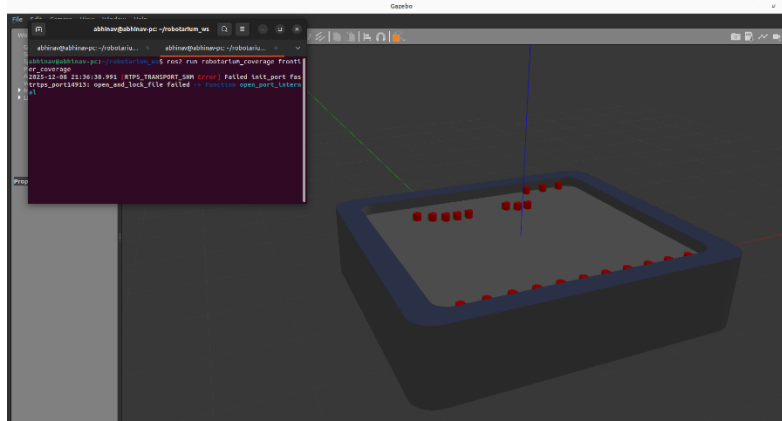


Figure 14. *Gazebo Simulation of 3D Frontier Exploration*

The frontier exploration algorithm provides consistent motion of the robots towards the boundary between the known and unknown regions within the given environment. Providing a spreading behavior as shown in Figure 11 where each robot independently identifies the frontier cells from its localized occupancy grid giving its ability to individually sense the nearest frontier target for a collaborative approach. This behavior confirms the theoretical gradient-driven frontier selection rule derived in Section 3, demonstrating attraction toward unexplored space and progressive expansion of the collective coverage. However, robots occasionally converge on similar frontier regions when their observations overlap, reproducing the non-unique equilibrium points predicted in the theoretical analysis and illustrating the algorithm's susceptibility to deadlock or temporary stagnation. Despite these limitations, inter-robot collisions do not occur, validating the safety guarantees of the distributed control architecture even under full 3D physics.

4.7.5 3D Voronoi Coverage

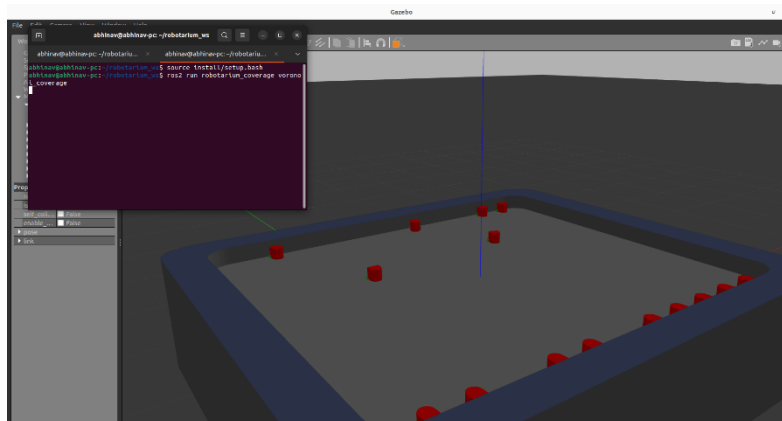


Figure 15. *3D Gazebo simulation of the Voronoi coverage algorithm. Robots disperse across the arena according to Voronoi partitioning, moving toward locally computed centroids.*

The Voronoi coverage algorithm provides consistent motion of distribution amongst robots within the given environment by repeatedly computing Voronoi partitions from their initial positions navigating towards the centroids of their respective cells as shown in Figure 12 providing consistent spatial separation. These findings are consistent with the theoretical principles of centroidal Voronoi tessellations established in Section 3. While there are several factors affecting the system that exist in three dimensions, such as latencies and the presence of sensor noise, the system will converge predictably to a balanced coverage configuration.

4.6 Hardware Validation



Figure 15. *Physical Mobile Robots Used for Hardware Validation. Differential-drive mobile robots are used for real-world validation of the PFBC controller. The platforms include onboard sensing modules (depth camera, LiDAR, or RGB-D), a multi-tier electronics stack for computation and communication, and a robust wheeled base enabling reliable ground motion*

The robots shown in Figure 15 represent research grade mobile platforms commonly used in multi-robot systems experiments, which is a Pheeno robots developed in Professor Berman's lab. This robot consists of a differential-drive base, onboard microcontrollers, and a sensing module that include RGB cameras and short-range depth sensors for localized sensing. The electronics shown in the back provides robot with wireless communication, and some of the feedback capabilities which could potentially serve the purpose of implementing distributed coverage algorithms.

Real robots have been developed as seen in figure 15 and these algorithms can be tested on a swarm of similar kind of robots for future scope of this project.

5.0 Verification (contributions: Arvind Kaushik)

Potential Field Based Coverage Controller:

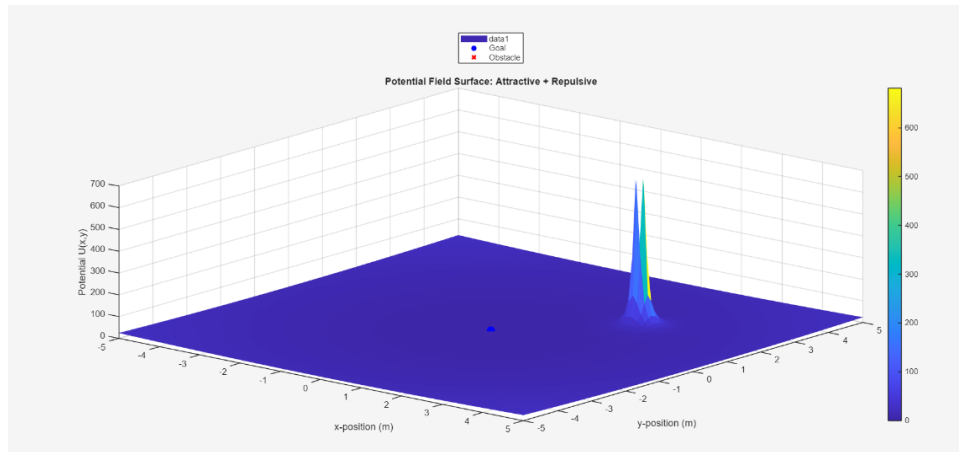


Figure 16. *Potential Field Surface: Plot showing attractive and repulsive field where robots follow the negative gradient of the surface to approach the goal.*

The graph shown above illustrates the 3D potential field surface for Potential Field Based Coverage algorithm. The surface represents potential function $\Phi(x,y)$ evaluated over the workspace, where robots navigate by following the negative gradient $\nabla\Phi$, moving from regions of high potential to low potential. Tall spike shown above is a visual representation of strong repulsive potential caused by a barrier (obstacle or a neighboring robot). This graph verifies differentiable potential field for stable motion, attractive global region as it guarantees the robots are directed towards their respective targets.

Voronoi based coverage:

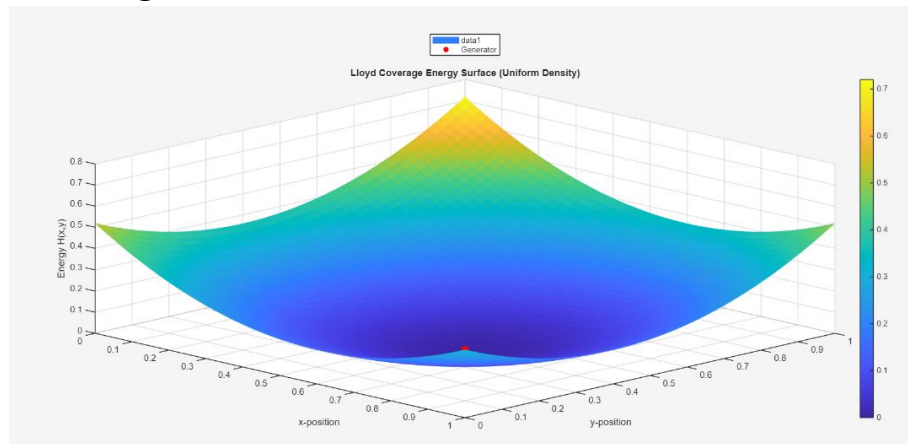


Figure 17. *Lloyd Coverage Energy Surface: square distance coverage energy of a single generator under uniform density where each Voronoi iteration move the generators towards the minima of the energy*

The graph shown above provides a visualization of the distance squared energy landscape which governs the behavior of Voronoi based coverage algorithm. The surface for this graph follows a smooth trend of convex basin representing the transitional energy increase as the inspector robot sequentially moves away from the centroid and decreases as it approaches the center region. This proves the theoretical formulation from section 3 of Lloyd Algorithm at the robots is expected to perform a gradient descent on the coverage energy driving towards the minimizer of its region cost.

Sweep based coverage:

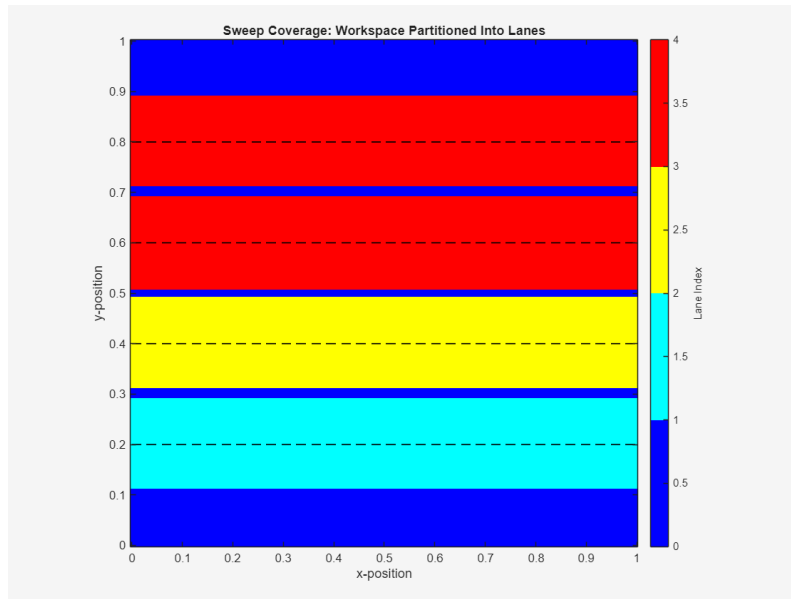


Figure 18. Robot assigned each lane performs back and forth sweeps within its lane for complete coverage

The sweep-based coverage plot indicates the given environment into horizontal parallel lanes as each inspector robot is assign to a particular lane to ensure deterministic and non-overlapping exploration. The representation of this graph verifies the characteristic of sweep coverage as hybrid automation with predictable transitions as mentioned in section 3 long the lane boundaries. The stricture among each lane confirms that the inspector robot remains in its confined area for exploration validating a collision free coordinated motion derived in theoretical analysis.

Frontier based exploration:

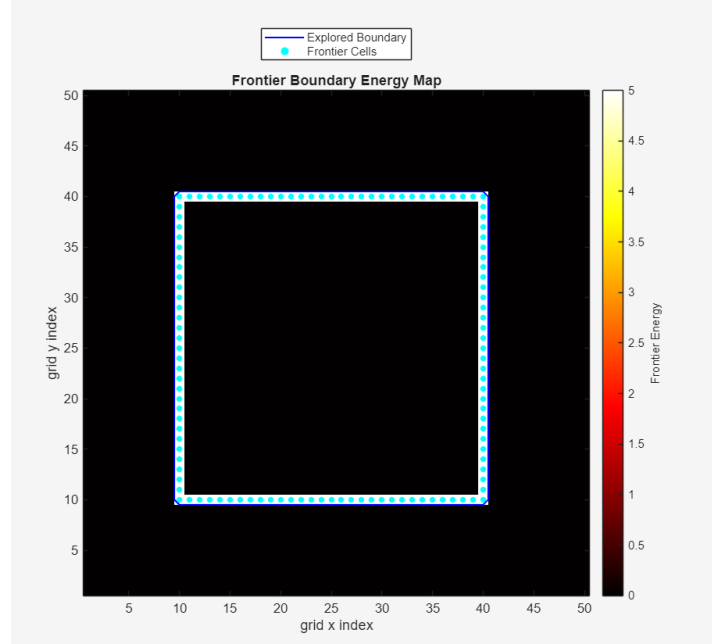


Figure 19. *The boundary between explored and unexplored radius corresponds to the bright colors guiding robots towards informative locations*

The frontier-based coverage exploration shown in Figure 19 represents the emergence of the informative gradient between known and unknown regions indicating the area of explored and unexplored. The energy band outlining the inner part of the graph represents the frontier cells within a given space. The immediate change of the frontier ring confirms that the algorithm successfully identifies the informative boundary set $\partial\Omega_U$, validating the second property regarding correct frontier detection. At the same time, the uniformity of the frontier ring illustrates the potential for multiple robots to be drawn toward similar frontier segments, consistent with the theoretical prediction of non-unique equilibria in symmetric environments.

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